# Lab 3 – Working with Git History, Stashing, and Reverting Commits

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**Course Name: Cloud DevOps**

**Section: B**

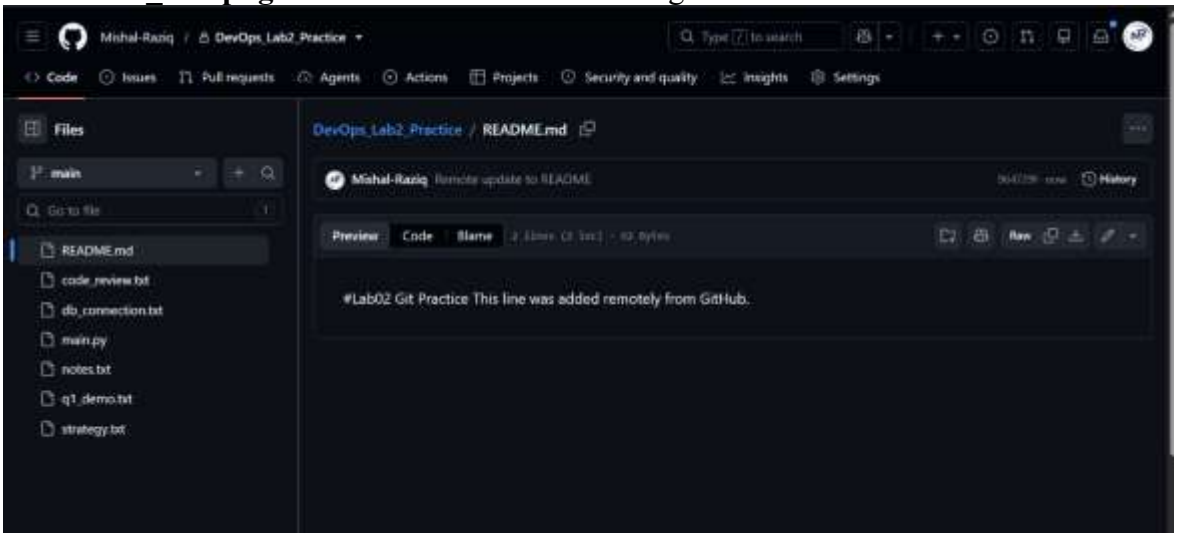
**GitHub username: mishal-raziq**

**Submitted to : Sir Waqas**

***Department of Computer Science***

**Task 1 – Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)**

- **remote_edit.png**: GitHub web interface showing the README file edited and committed remotely.




- **local_commit.png**: Terminal showing the local git commit for README edits before pulling.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
#Lab02 Git Practice
this line was added locally.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "local update to READEME"
[main 5ead1a4] local update to READEME
 1 file changed, 1 insertion(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **push_error.png**: Terminal output showing the rejected git push due to remote changes.

```
 1 file changed, 1 insertion(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:mishalraziq/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **merge_commit.png**: Terminal showing a non-rebase pull creating a merge commit.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git pull --no-rebase origin main
From github.com:mishalraziq/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "merge commit done"
[main 01eb22f] merge commit done
```

- **push_after_merge.png**: Terminal displaying successful push after merging remote changes.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 918 bytes | 131.00 KiB/s, done.
Total 9 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
To github.com:mishalraziq/DevOps_Lab2_Practice.git
   964729f..01eb22f  main -> main
```

- **rebase_pull.png**: Terminal output showing a clean linear history update via git pull --rebase.

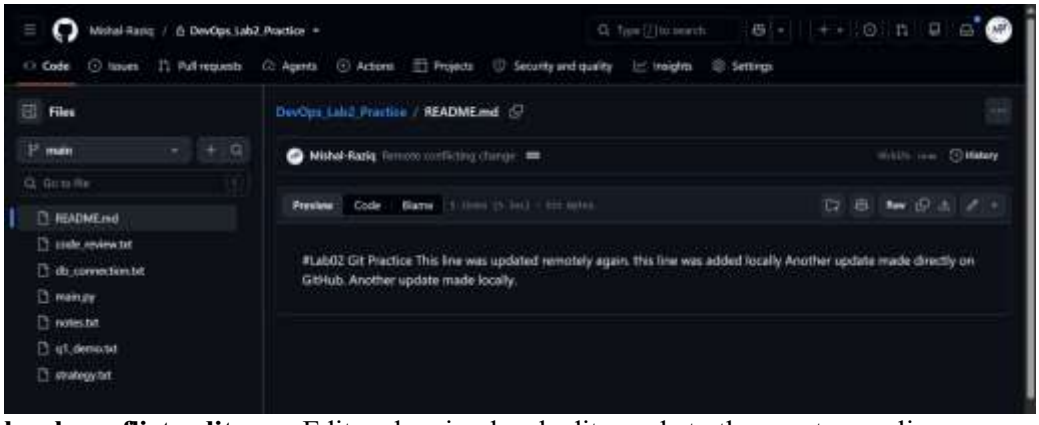
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1022 bytes | 511.00 KiB/s, done.
From github.com:mishalraziq/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   01eb22f..0d96e38  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 822a8ee... second local update
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 822a8ee... # second local update
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git rebase --continue
[detached HEAD ad00afe] second local update
 1 file changed, 1 insertion(+)
Successfully rebased and updated refs/heads/main.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **push_after_rebase.png**: Terminal displaying successful push following the rebase operation.

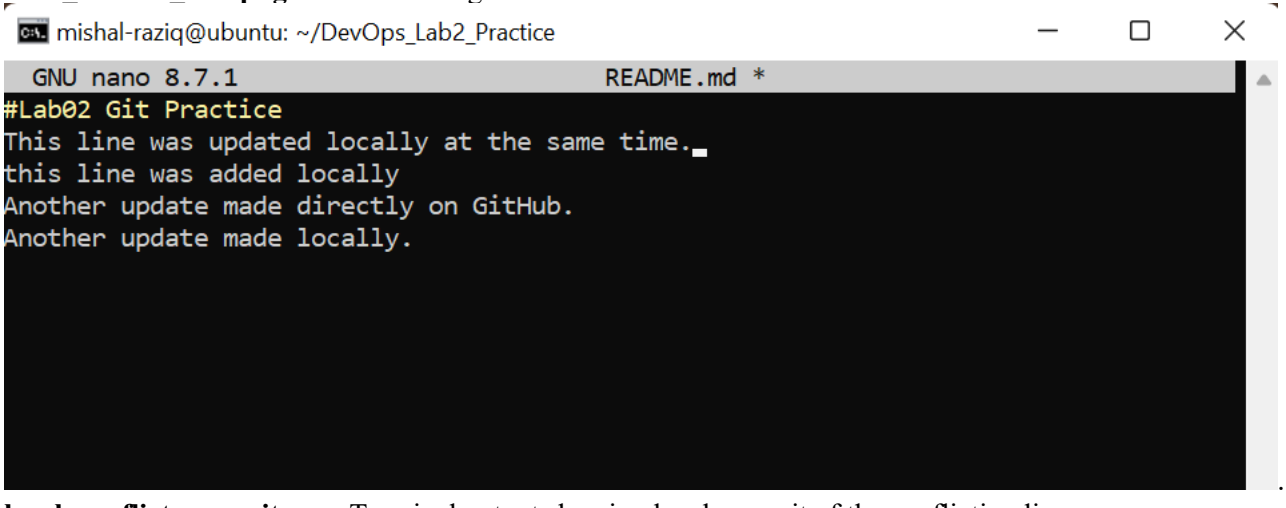
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 337 bytes | 112.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
To github.com:mishalraziq/DevOps_Lab2_Practice.git
   0d96e38..ad00afe  main -> main
branch 'main' set up to track 'origin/main'.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

**Task 2 – Creating and Resolving Merge Conflicts Manually**

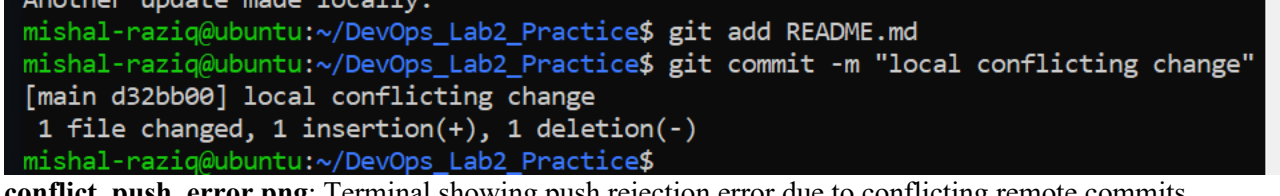
- **remote_conflict_edit.png**: GitHub web page showing a conflicting edit saved on the remote branch.



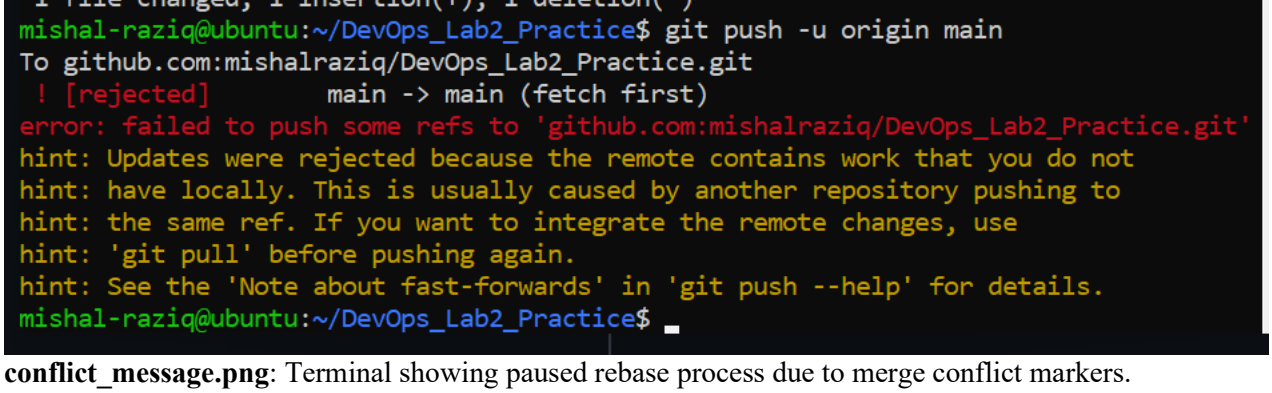

- **local_conflict_edit.png**: Editor showing local edits made to the exact same line as remote




- **local_conflict_commit.png**: Terminal output showing local commit of the conflicting line.




- **conflict_push_error.png**: Terminal showing push rejection error due to conflicting remote commits.




- **conflict_message.png**: Terminal showing paused rebase process due to merge conflict markers.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1014 bytes | 253.00 KiB/s, done.
From github.com:mishalraziq/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   ad00afe..8fcfd2b  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply d32bb00... local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abor
t".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply d32bb00... # local conflicting change
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **resolved_readme.png**: Editor view showing README with conflict markers manually resolved and cleaned.

```
mishal-raziq@ubuntu: ~/DevOps_Lab2_Practice
#Lab02 Git Practice
<<<<<<< HEAD
This line was updated remotely again and This line was updated locally at the same time
this line was added locally
Another update made directly on GitHub.
Another update made locally.
```

- **rebase_continue.png**: Terminal displaying git rebase --continue execution after staging the fix.

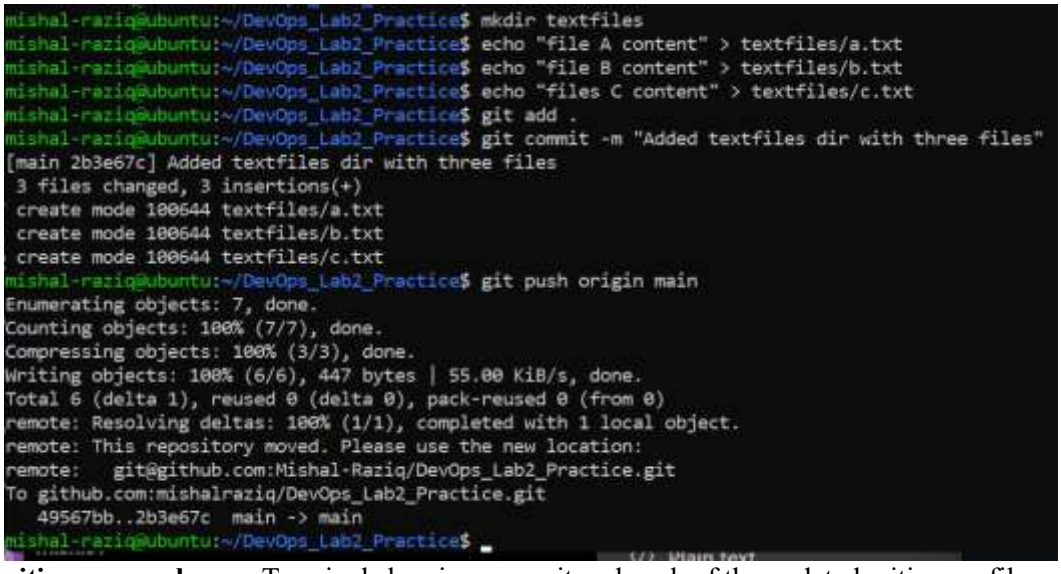
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git rebase --continue
[detached HEAD 49567bb] local conflicting change
 1 file changed, 2 insertions(+), 1 deletion(-)
Successfully rebased and updated refs/heads/main.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **push_after_resolve.png**: Terminal confirming successful push after manually resolving the merge conflict.

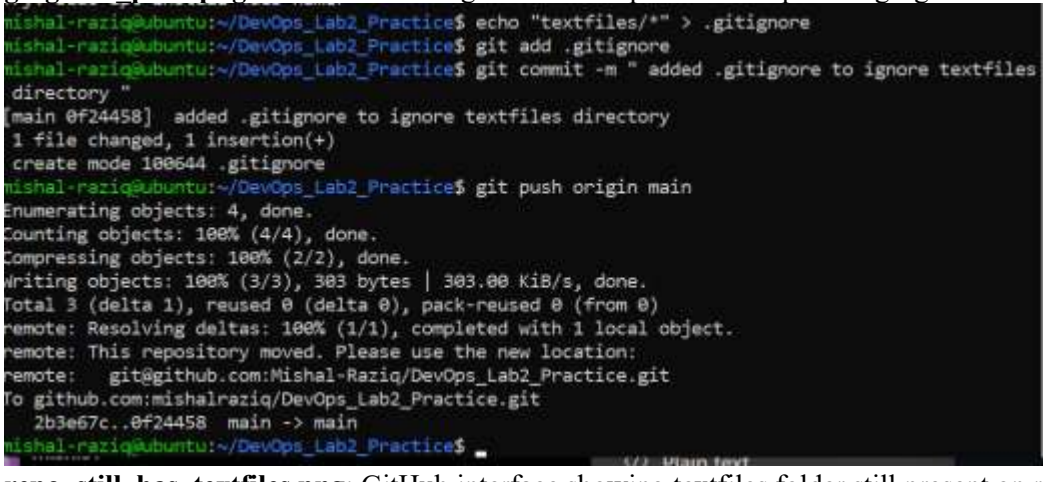
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 412 bytes | 137.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
To github.com:mishalraziq/DevOps_Lab2_Practice.git
   8fcfd2b..49567bb  main -> main
branch 'main' set up to track 'origin/main'.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

**Task 3 – Managing Ignored Files with .gitignore and Removing Tracked Files**

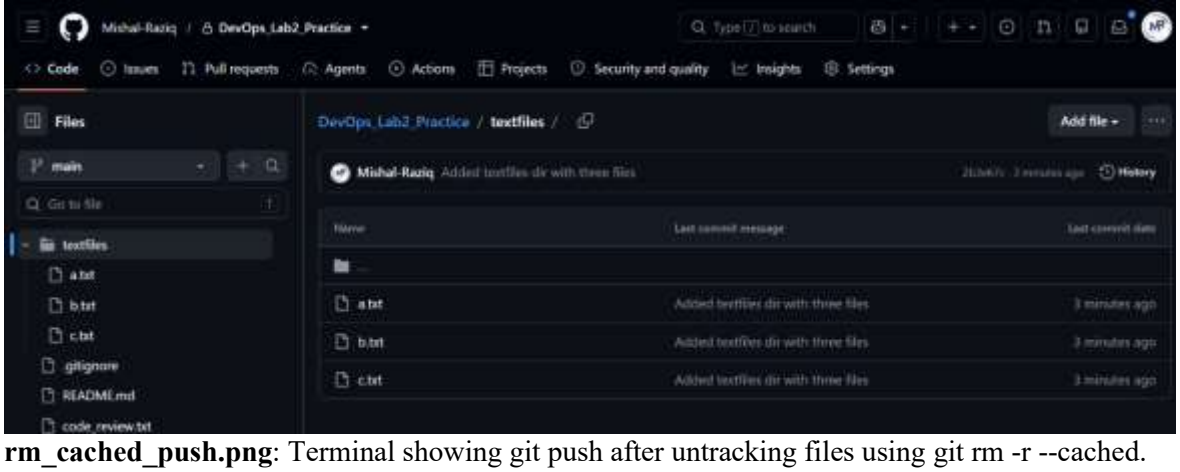
- **push_textfiles.png**: Terminal confirming push of the newly added textfiles directory.




- **gitignore_push.png**: Terminal showing commit and push of the updated .gitignore file.



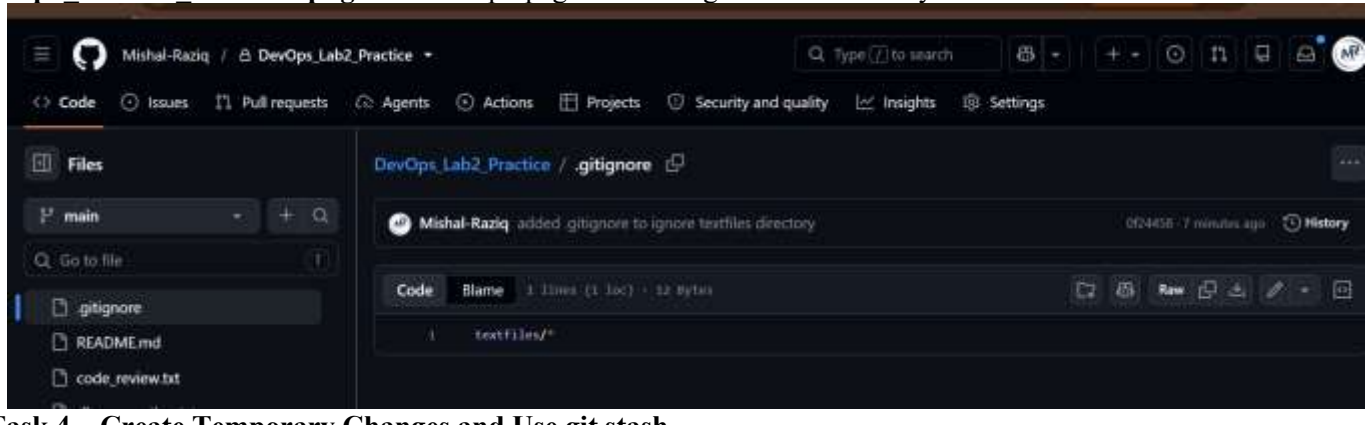

- **repo_still_has_textfiles.png**: GitHub interface showing textfiles folder still present on remote.




- **rm_cached_push.png**: Terminal showing git push after untracking files using git rm -r --cached.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m " removed tracked textfiles
 dir "
[main 32c2e56]  removed tracked textfiles dir
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 233 bytes | 233.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
To github.com:mishalraziq/DevOps_Lab2_Practice.git
   0f24458..32c2e56  main -> main
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **repo_textfiles_removed.png**: GitHub repo page confirming textfiles directory is removed from remote.




**Task 4 – Create Temporary Changes and Use git stash**

- **modified_readme.png**: Terminal output showing commit of initial README modifications on feature branch.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano READMEA.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m " change the readme file"
[feature-branch 8173f3a]  change the readme file
 1 file changed, 1 insertion(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **checkout_error.png**: Terminal displaying error when attempting branch switch with uncommitted changes.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **stash_command.png**: Terminal confirming successful execution of git stash to store changes.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git stash
Saved working directory and index state WIP on feature-branch: 8173f3a  change the re
adme file
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **branch_switched.png**: Terminal showing successful checkout to main branch after stashing.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git stash
Saved working directory and index state WIP on feature-branch: 8173f3a  change the re
adme file
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **back_to_feature.png**: Terminal showing checkout back to feature-branch.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **status_clean.png**: Terminal showing git status reporting a clean working directory

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
```

- **stash_pop.png**: Terminal displaying restoration of stashed changes via git stash pop.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (27f540d3fb1453dd523eef83e997e13d6fa684af)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- 

**Task 5 – Checkout a Specific Commit Using git log**

- **log_before_checkout.png**: Terminal output of git log --oneline listing past commit hashes.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
8173f3a (HEAD -> feature-branch)  change the readme file
32c2e56 (origin/main, origin/HEAD)  removed tracked textfiles dir
0f24458  added .gitignore to ignore textfiles directory
2b3e67c Added textfiles dir with three files
49567bb local conflicting change
8fcfd2b Remote conflicting change
ed00afe second local update
0d96e38 Second remote README update
01eb22f merge commit done
4ea41a7 merge commit done
5ead1a4 local update to READEME
964729f Remote update to README
6d1868d (origin/feature/q1-demo, origin/feature/db-connection, feature/q1-demo, feature/db-connection) added
3cc5caa Merge pull request #2 from Mishal-Raziq/feature/code-review
11b7f14 (feature/code-review) added
6f682c8 added
04b3211 added
f9a0f86 (origin/staging, origin/develop, staging, develop) added
4d16375 (origin/feature/strategy-demo, feature/strategy-demo)  updated
eacbc43 add
10754cb Merge pull request #1 from mishalraziq/feature/db-connection
a1dd4ef add feature
bfd84d3 (origin/feature-1, feature-1) ADD NEW FUNCTION TO main.py
ce790f8 updated
3f4a791 ADD notes.txt
fb244bd (origin/bugfix/user-auth-error) Initial Commit
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **detached_head.png**: Terminal message showing repository in a 'detached HEAD' state after checking out a hash.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout 8173f3a
M       README.md
Note: switching to '8173f3a'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 8173f3a  change the readme file
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **back_to_main.png**: Terminal confirming switch back to main branch from detached HEAD.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

**Task 6 – Resetting Commits (Soft vs Hard Reset)**

- **first_commit.png**: Terminal output confirming creation of the first test commit.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add .
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "first commit"
[main ba78e73] first commit
 1 file changed, 1 insertion(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **second_commit.png**: Terminal output showing the second test commit creation.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add .
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "second commit"
[main b28e954] second commit
 1 file changed, 1 insertion(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **log_before_reset.png**: Terminal output of git log --oneline displaying both commits before reset.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
b28e954 (HEAD -> main) second commit
ba78e73 first commit
92edd6e done
32c2e56 (origin/main, origin/HEAD)  removed tracked textfiles dir
0f24458  added .gitignore to ignore textfiles directory
2b3e67c Added textfiles dir with three files
749567bb local conflicting change
8fcfd2b Remote conflicting change
sd00afe second local update
0d96e38 Second remote README update
01ab22f merge commit done
4ea41a7 merge commit done
5ead1a4 local update to README
964729f Remote update to README
6d1868d (origin/feature/q1-demo, origin/feature/db-connection, feature/q1-demo, feature/db-connection) added
3cc5caa Merge pull request #2 from Mishal-Raziq/feature/code-review
11b7f14 (feature/code-review) added
6f682c8 added
04b3211 added
f9a0f86 (origin/staging, origin/develop, staging, develop) added
4d16375 (origin/feature/strategy-demo, feature/strategy-demo)  updated
eacbc43 add
10754cb Merge pull request #1 from mishalraziq/feature/db-connection
a1dd4ef add feature
bfd84d3 (origin/feature-1, feature-1) ADD NEW FUNCTION TO main.py
ce790f8 updated
3f4a791 ADD notes.txt
fb244bd (origin/bugfix/user-auth-error) Initial Commit
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

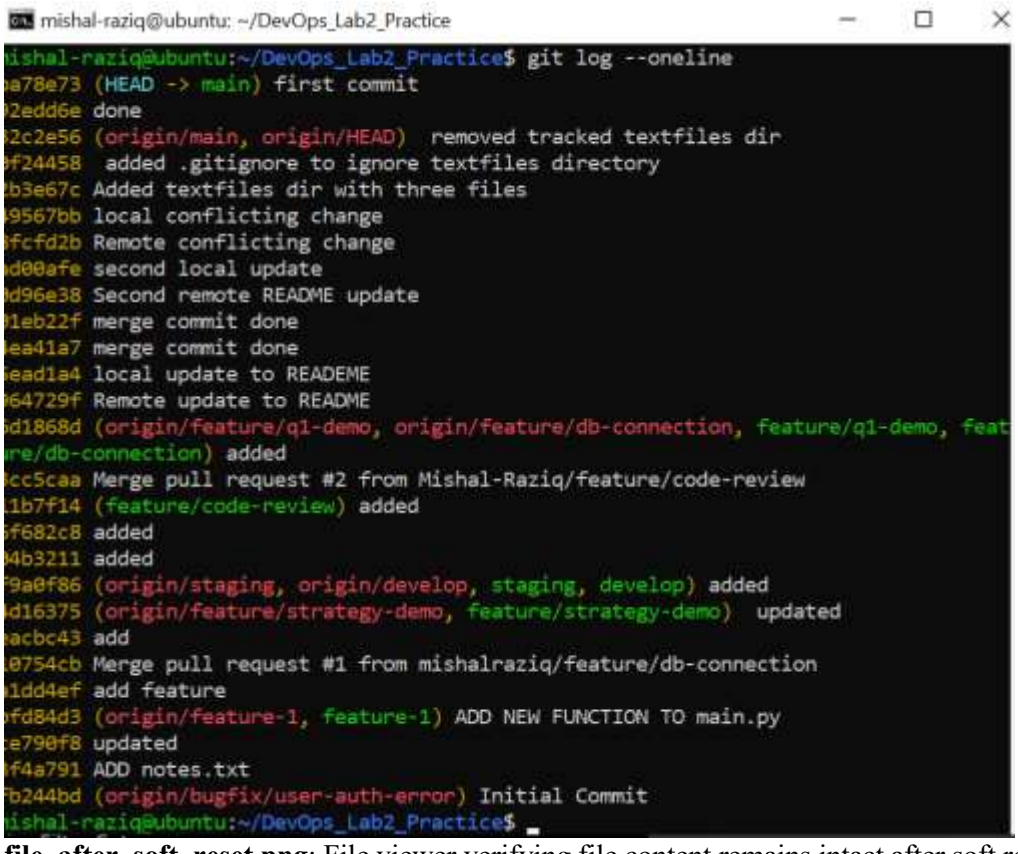
- **file_before_reset.png**: File viewer confirming presence of both commit lines in the file.

```
mishal-raziq@ubuntu: ~/DevOps_Lab2_Practice
  GNU nano 8.7.1                        README.md
#Lab02 Git Practice
<<<<<<< HEAD
This line was updated remotely again and This line was updated locally at the same time.
this line was added locally
Another update made directly on GitHub.
Another update made locally.
line is added for resetting commits.
second commit line.
```

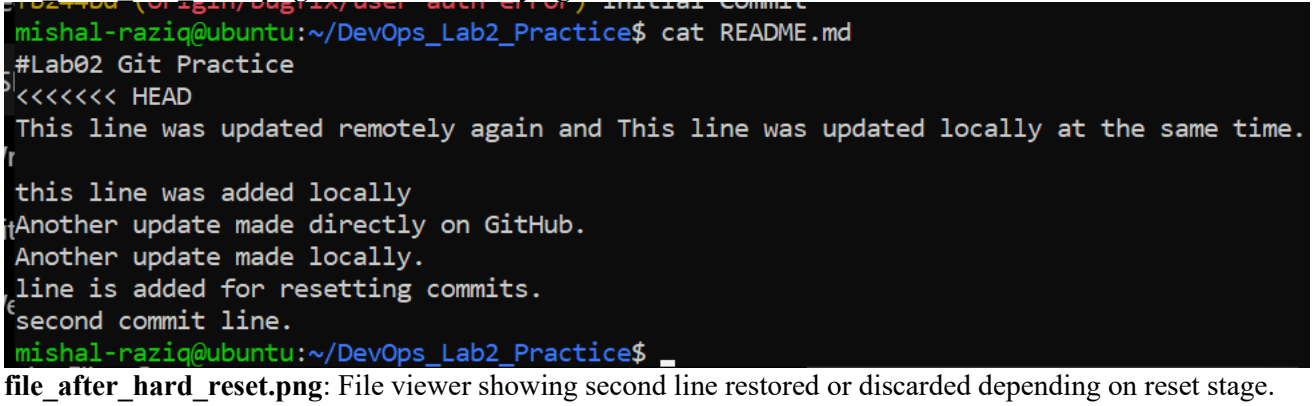
- **soft_reset.png**: Terminal output demonstrating execution of git reset --soft HEAD~1.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git reset --soft HEAD~1
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

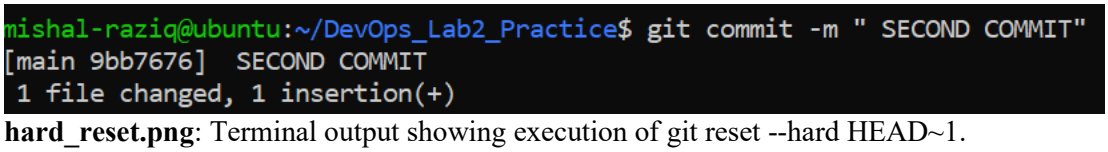
- **log_after_soft_reset.png**: Terminal log confirming commit history rolled back while changes remain.



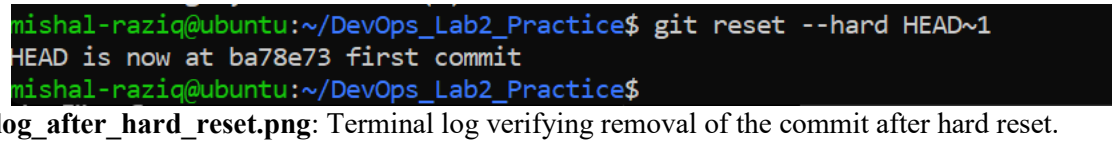

- **file_after_soft_reset.png**: File viewer verifying file content remains intact after soft reset.




- **file_after_hard_reset.png**: File viewer showing second line restored or discarded depending on reset stage.




- **hard_reset.png**: Terminal output showing execution of git reset --hard HEAD~1.




- **log_after_hard_reset.png**: Terminal log verifying removal of the commit after hard reset.

```
mishal-raziq@ubuntu: ~/DevOps_Lab2_Practice
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline
ba78e73 (HEAD -> main) first commit
92edd6e done
32c2e56 (origin/main, origin/HEAD)  removed tracked textfiles dir
8f24458  added .gitignore to ignore textfiles directory
2b3e67c Added textfiles dir with three files
49567bb local conflicting change
8fcfd2b Remote conflicting change
ad00afe second local update
0d96e38 Second remote README update
81eb22f merge commit done
4ea41a7 merge commit done
5ead1a4 local update to READEME
964729f Remote update to README
6d1868d (origin/feature/q1-demo, origin/feature/db-connection, feature/q1-demo, feature/
db-connection) added
3cc5caa Merge pull request #2 from Mishal-Raziq/feature/code-review
11b7f14 (feature/code-review) added
6f682c8 added
04b3211 added
f9a0f86 (origin/staging, origin/develop, staging, develop) added
4d16375 (origin/feature/strategy-demo, feature/strategy-demo)  updated
eacbc43 add
10754cb Merge pull request #1 from mishalraziq/feature/db-connection
a1dd4ef add feature
bfd84d3 (origin/feature-1, feature-1) ADD NEW FUNCTION TO main.py
ce790f8 updated
3f4a791 ADD notes.txt
fb244bd (origin/bugfix/user-auth-error) Initial Commit
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **file_after_hard_reset.png**: File viewer confirming complete removal of changes after hard reset.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ cat README.md
#Lab02 Git Practice
<<<<<<< HEAD
This line was updated remotely again and This line was updated locally at the same time.
this line was added locally
Another update made directly on GitHub.
Another update made locally.
line is added for resetting commits.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

**Task 7 – Amending the Last Commit**

- **first_amend_commit.png**: Terminal showing initial commit prior to making an amend edit.

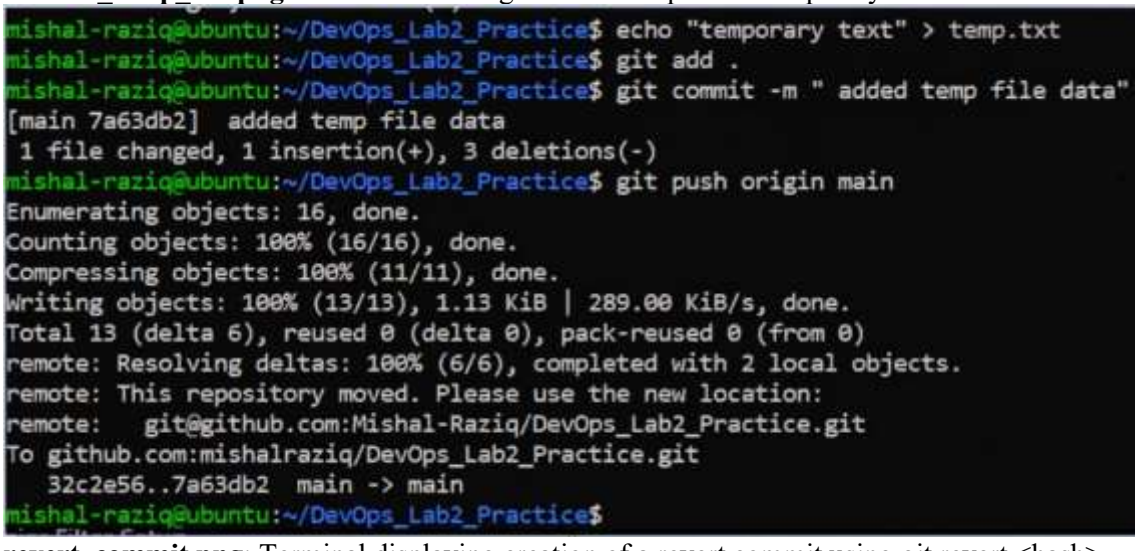
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ nano README.md
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add .
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "fix log message"
[main dedab54] fix log message
 1 file changed, 1 insertion(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **amend_commit.png**: Terminal confirming updated commit details using git commit --amend.

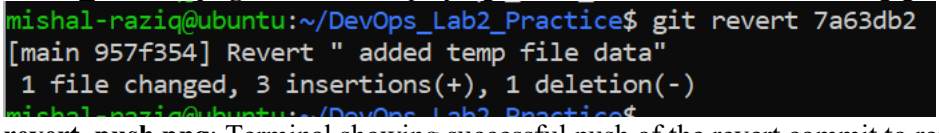
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ echo "additional update for l
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add .
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit --amend
[main 91b48ad] fix log message
 Date: Sat Oct 17 09:50:27 2026 +0000
 2 files changed, 2 insertions(+)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

**Task 8 – Reverting a Commit (Safe Undo on Remote Branch)**

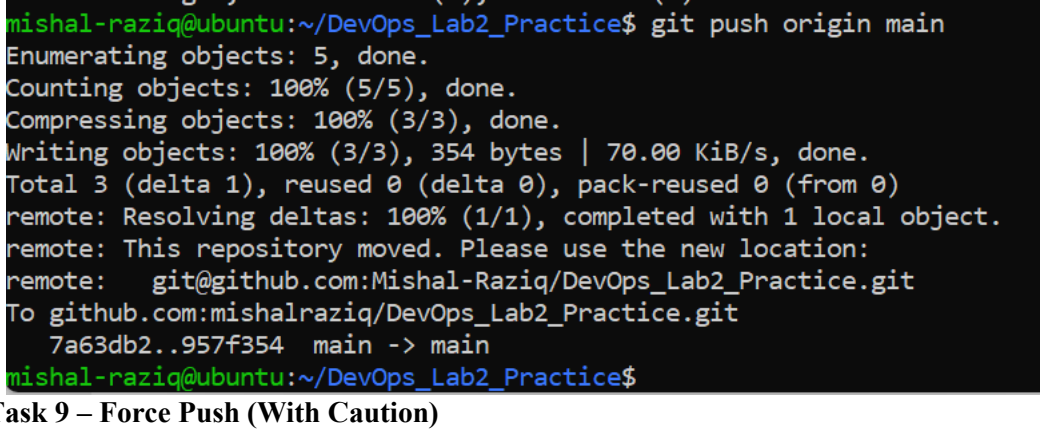
- **commit_temp_file.png**: Terminal showing commit and push of temporary file to remote branch.




- **revert_commit.png**: Terminal displaying creation of a revert commit using git revert <hash>.

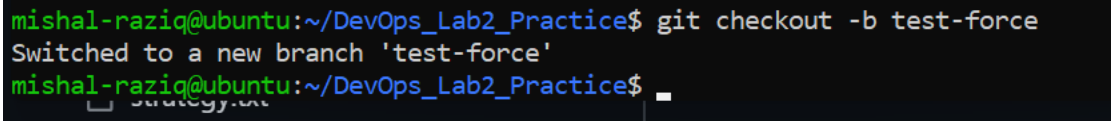



- **revert_push.png**: Terminal showing successful push of the revert commit to remote repository.

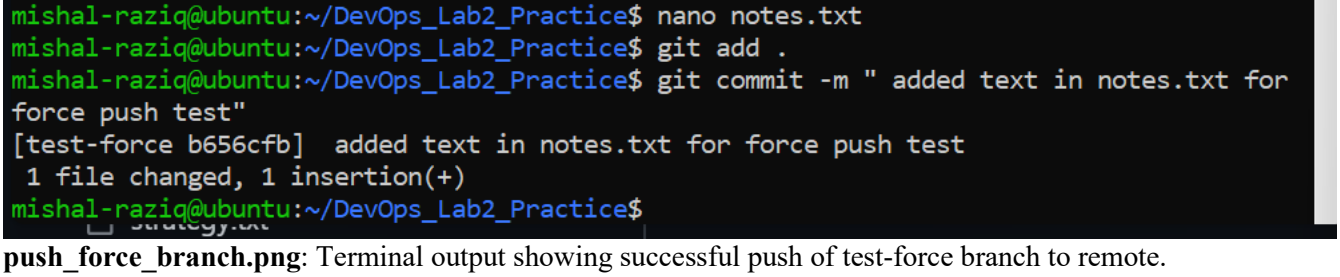



**Task 9 – Force Push (With Caution)**

- **new_branch.png**: Terminal confirming creation and checkout of test-force branch.




- **force_commit.png**: Terminal showing a test commit made on test-force branch.




- **push_force_branch.png**: Terminal output showing successful push of test-force branch to remote.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 662 bytes | 94.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/Mishal-Raziq/DevOps_Lab2_Practice/pull/new/test-force
remote:
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 * [new branch]      test-force -> test-force
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **hard_reset_force.png**: Terminal output showing local hard reset execution on test-force.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git reset --hard HEAD~1
HEAD is now at 8173f3a  change the readme file
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **normal_push.png**: Terminal showing standard push rejected due to non-fast-forward state.

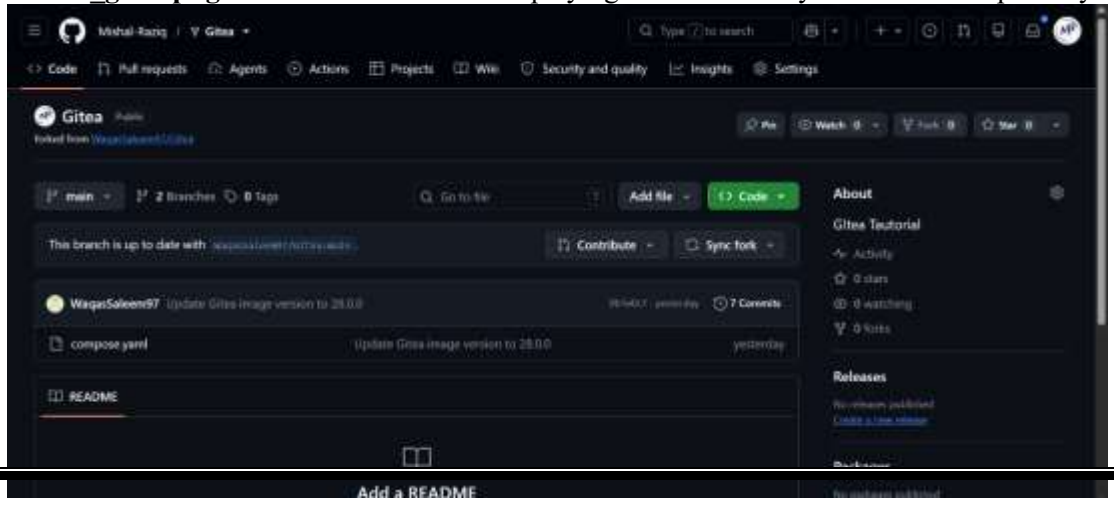
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:mishalraziq/DevOps_Lab2_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

- **force_push.png**: Terminal confirming force push completion using git push origin test-force --force.

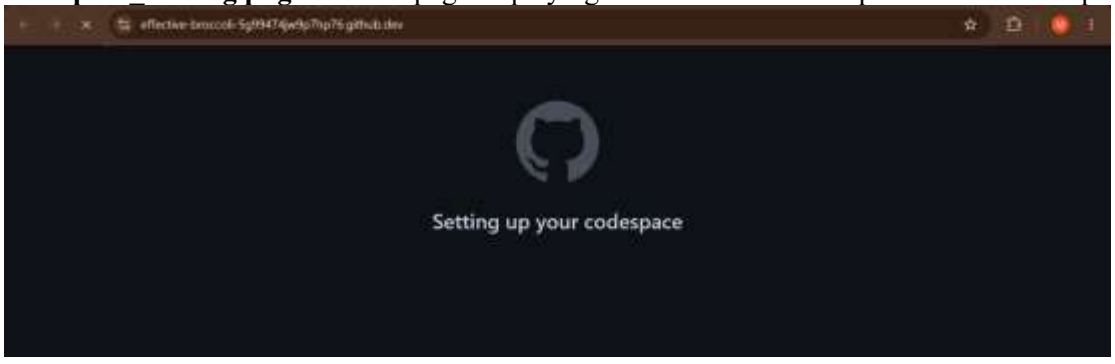
```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin test-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 + b656cfb...8173f3a test-force -> test-force (forced update)
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

**Task 10 – Running Gitea in GitHub Codespaces via Docker Compose**

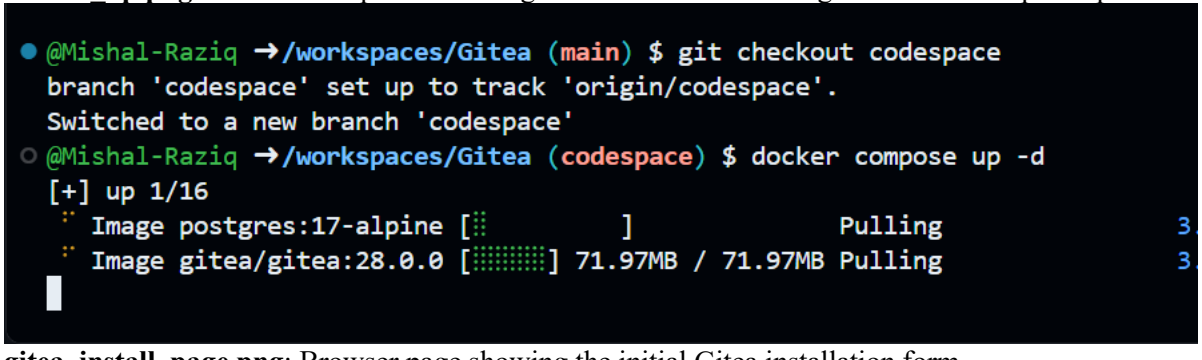
- **forked_gitea.png**: GitHub browser view displaying the successfully forked Gitea repository.



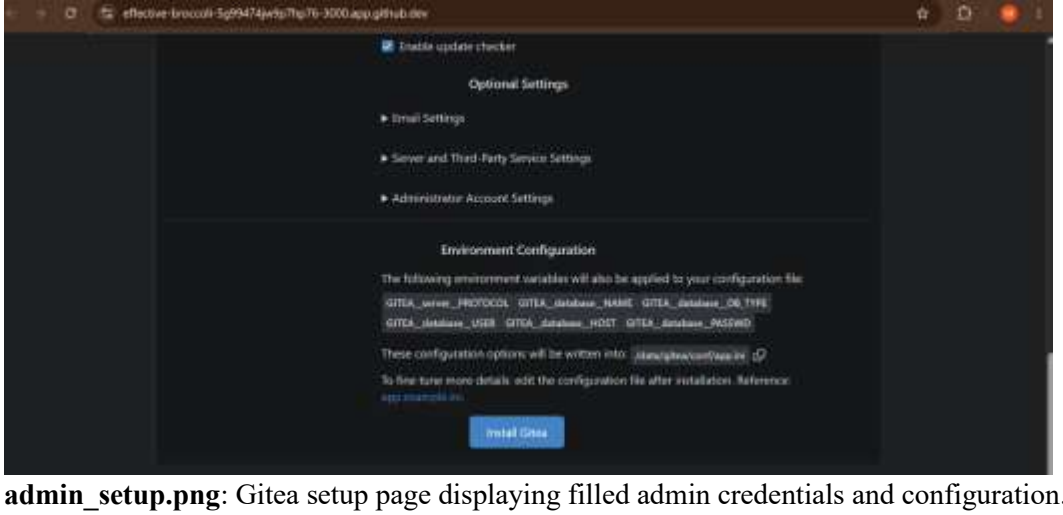
- **codespace_loading.png**: Browser page displaying initialization and setup of GitHub Codespace.



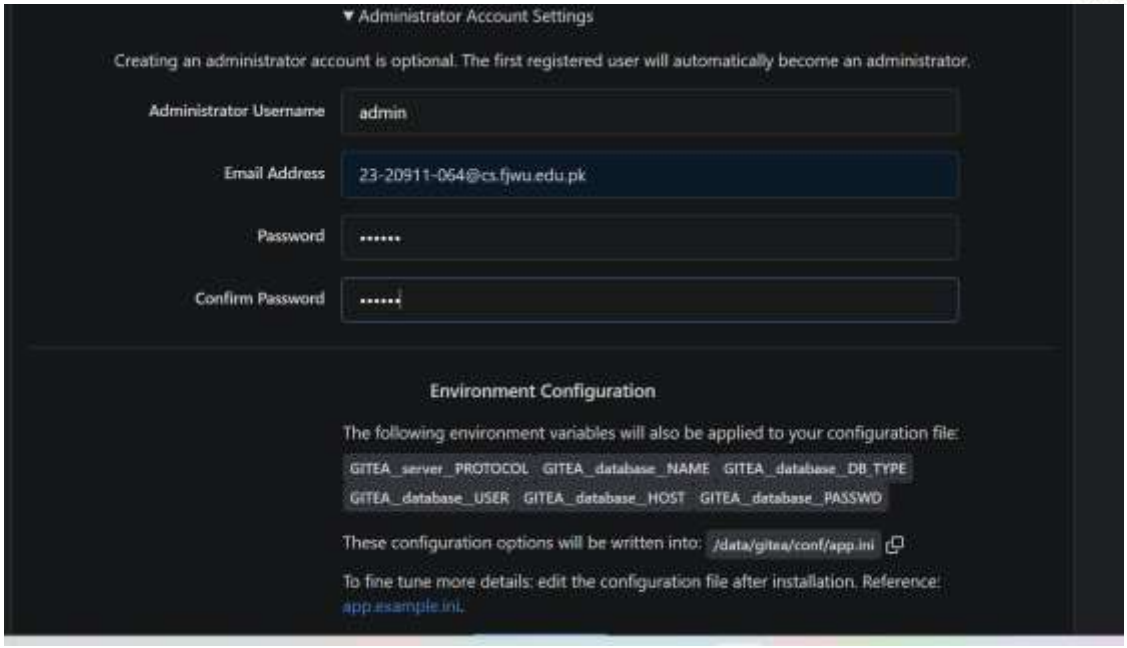

- **docker_up.png**: Terminal output confirming Gitea containers running via docker compose up -d.



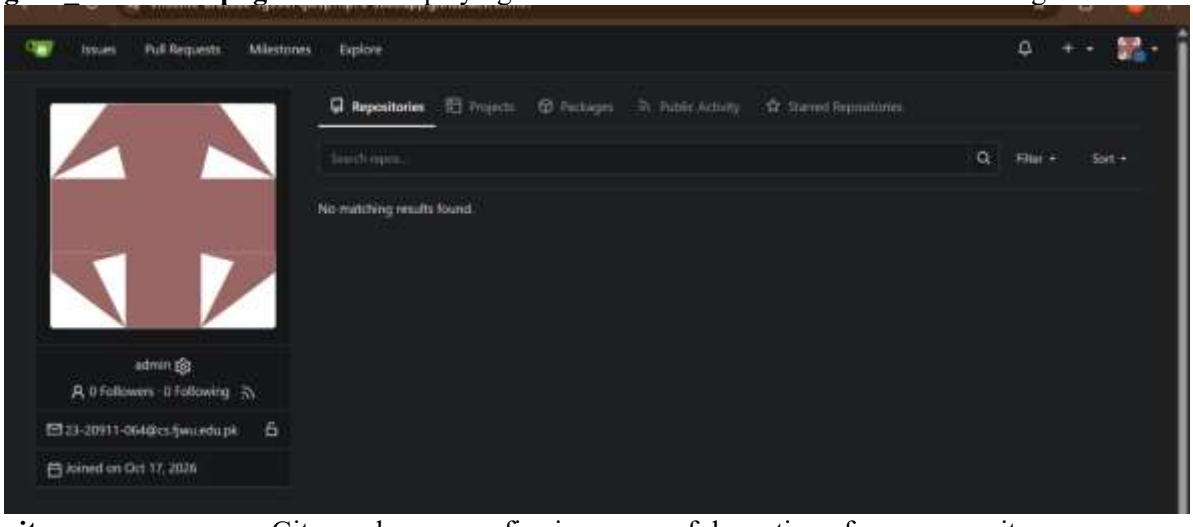

- **gitea_install_page.png**: Browser page showing the initial Gitea installation form.



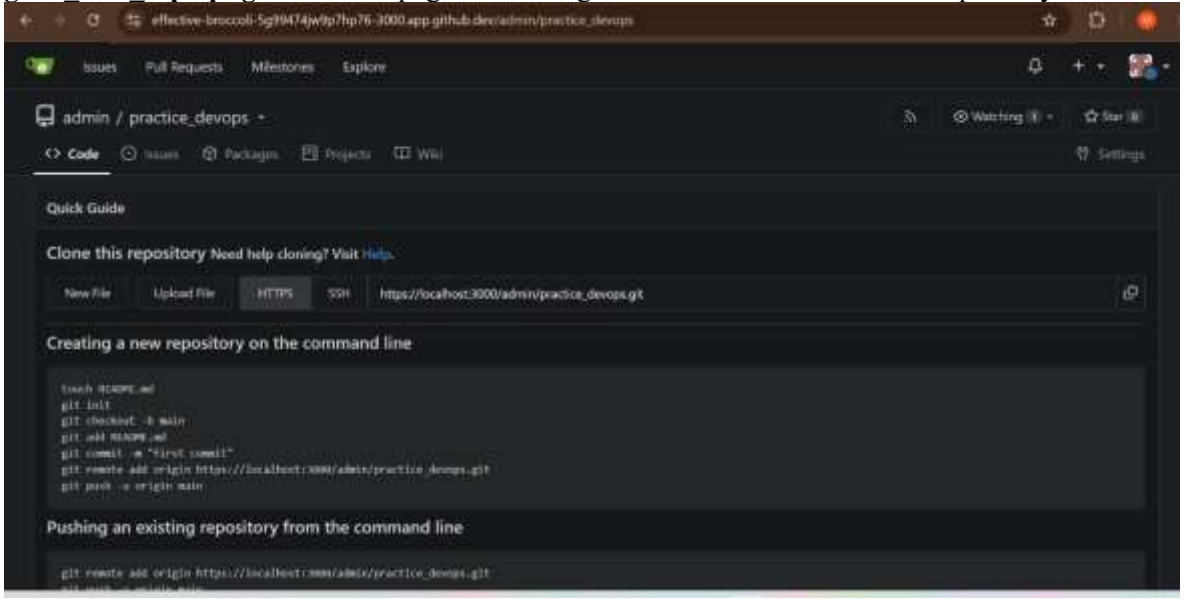

- **admin_setup.png**: Gitea setup page displaying filled admin credentials and configuration.




- **gitea_dashboard.png**: Browser displaying main Gitea dashboard after administrator login.

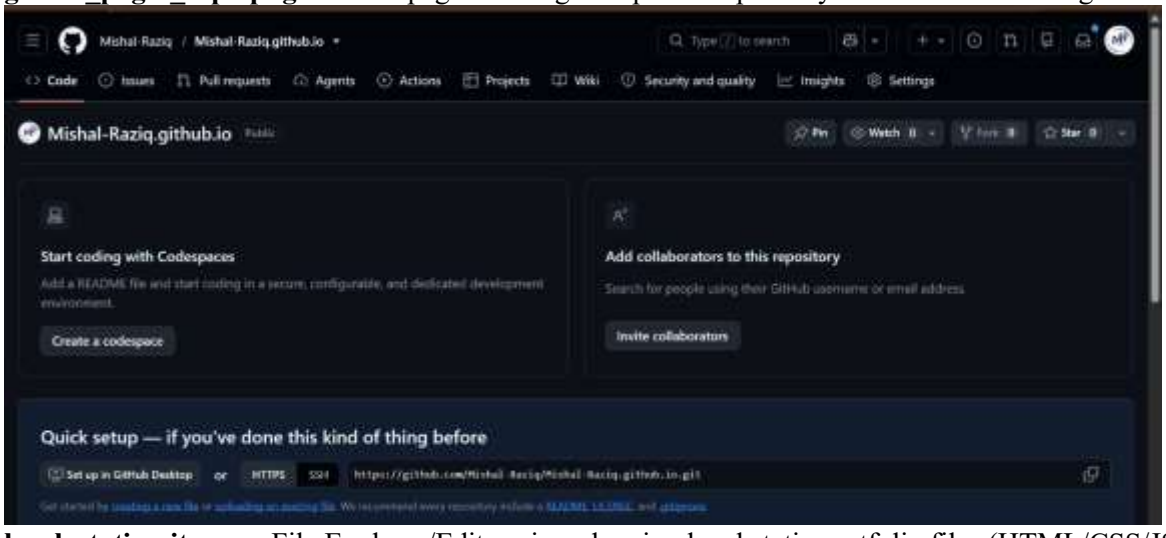



- **gitea_new_repo.png**: Gitea web page confirming successful creation of a new repository.



**Task 11 – Creating a GitHub Pages Portfolio Site**

- **github_pages_repo.png**: GitHub page showing new public repository named <username>.github.io.



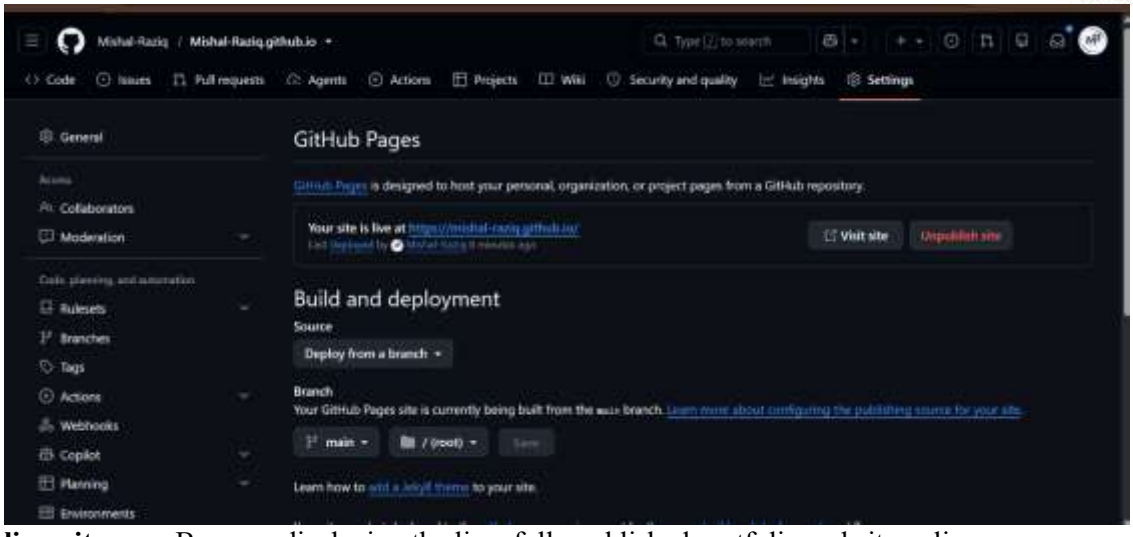

- **local_static_site.png**: File Explorer/Editor view showing local static portfolio files (HTML/CSS/JS).

```
mishal-raziq@ubuntu:~$ cd ~/Mishal-Raziq.github.io
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io$ mkdir my-portfolio
mkdir: my-portfolio: File exists
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io$ cd my-portfolio
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$ nano index.html
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$ ls
index.html
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$
```

- **push_static_site.png**: Terminal output confirming successful code push to GitHub Pages repo.

```
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$ git add .
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$ git commit -m "add port
folio site for github pages"
[master (root-commit) 8de80cb] add portfolio site for github pages
 1 file changed, 556 insertions(+)
 create mode 100644 index.html
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$ git push -u origin main
push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 3.66 KiB | 1.22 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Mishal-Raziq/Mishal-Raziq.github.io
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
mishal-raziq@ubuntu:~/Mishal-Raziq.github.io/my-portfolio$
```

- **github_pages_settings.png**: GitHub repository settings page displaying active GitHub Pages deployment link.




- **live_site.png**: Browser displaying the live, fully published portfolio website online.

